



BUSINESS BLUEPRINT

AOP AUTOMATION FOR NMD EOEG PLANT

Honeywell

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Business Blueprint

AOP Automation for NMD EOEG Plant

Prepared for:

Reliance Industries Ltd

Submitted by:

Honeywell Automation India Ltd (“Honeywell”)



Revision History

Rev.	Date	Originated By	Description of status and changes
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1. Introduction

The purpose of this document is to capture the detailed specifications for the Digitalized Annual Operating Plan (AOP) for NMD EOEG plant for NMD site.

This document is based on the Technical Proposal submitted by Honeywell to Reliance. It also refers to data collected during the workshop conducted at RCP with various MEG plant users across RIL sites.

1.1 Executive Summary

Reliance Industries Limited is India's leading private sector organization operating in various business verticals. The manufacturing sector is one such business vertical. RIL owns and operates multiple plants across various sites.

Defining and finalizing the Annual Operating Plan is a critical annual business process carried out by different functional teams. At present, this activity is carried out manually.

A Rev-0 budget is usually completed in Nov / Dec for the succeeding Financial Year (FY) and reviewed. The comments are addressed with updated information from the Business, as the next revised budget is expected by the end of January. Further revisions (if any) are incorporated in Feb / March to reflect updated business input as applicable.

The entries from the final rev of the budget are finalized in BPC before the beginning of the succeeding FY.

RIL aims to automate the AOP preparation process using a Data-Driven Model. This model leverages the last five years of AOP data from MIIS and incorporates plant-specific business rules (such as plant availability, running hours, feedstock availability, seasonal impact on production, etc.) to derive AOP figures in the expected formats.

As part of digitalization, Honeywell will support building a platform that will automate the Annual Operations Planning process for MEG technology.

1.2 Reference Documents

Document ID	Revision	Title	Document
NMD EOEG DCT	-	Digital AOP NMD EOEG DCT	 Digital AOP NMD EOEG DCT.xlsx
UAT Observation list		Digital AOP MEG UAT Observation list	 Digital AOP MEG UAT Observations.xl
EOE effective Operating Hours calculation		RIL AOP - MEG Maintenance Screen - Slowdown Logic_R&T Validated	 RIL AOP - MEG Maintenance Screen

Table 1 Reference Documents

2. AOP Design Basis

The table below captures different functions for the scope of AOP digitization design –

Sr. No.	Functions	Covered in Document
1.	Health, Safety and Environment (HSE)	For later release, post review and discussion
2.	Process/Operations	Covered
3.	Reliability and Maintenance	For later release, post review and discussion

Table 2 AOP Functional Scope

Function-wise, the detailed design basis is captured in the sections below.

2.1 Process and Operations Function

In this section, the budgeted Production and Norms for a plant are derived, using a standard approach as briefed in the sections below for the following criteria

- Budget Production/Production AOP
- Norms/Consumption AOP

Steps to generate the Production and Consumption AOP are captured in the below flow diagram.

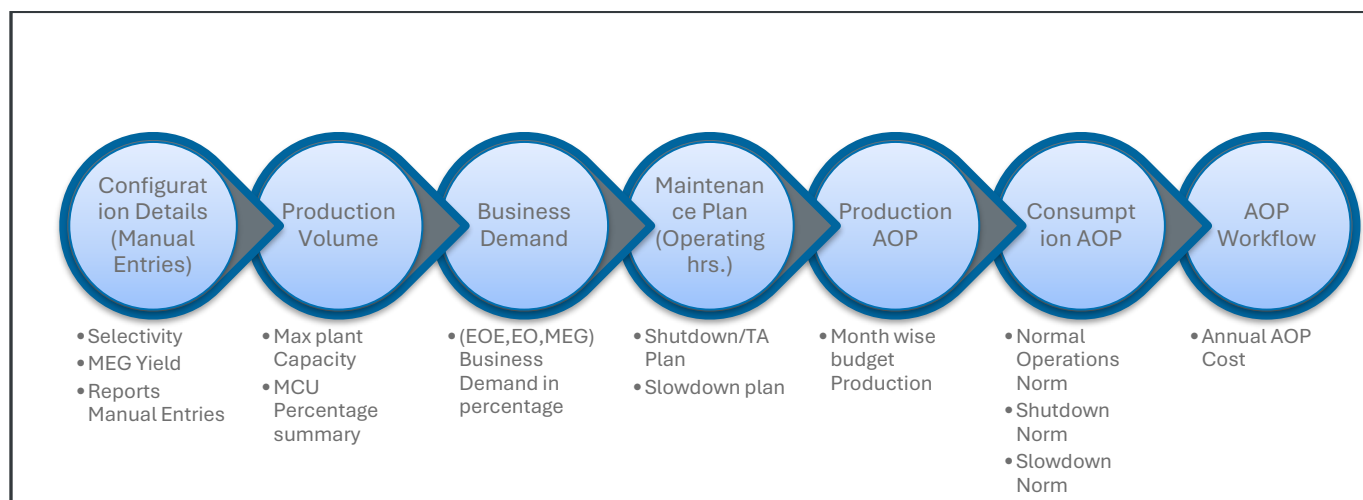


Figure 1 Digitized AOP Flow Diagram

2.1.1 Budget Production/Production AOP

Below are the steps captured to derive the budgeted AOP production values.

2.1.1.1 Configuration

The configuration screen serves as the foundation for the digital AOP process, allowing users to input essential data for production and consumption AOP calculations.

Plant-specific data, including Yield, Selectivity, and Consumption, are configured under the Configuration screen to support Production and Consumption Norms calculations and reports generation.

By default, the values entered for the current year will be automatically carried forward to the next year.

The user must insert the TST value month wise and save it with a valid reason. This will provide the Latent heat against TST.

NOTE- Latent heat will be calculated only if the user provides the TST temperature; otherwise, the latent heat value will be set to 0.

In configuration screen, below additional features and tabs are provided.

- AOP historical period basis added where Start and End date is provided to select historical period to calculate MCU and Norms data for selected period by default 05 years data will be loaded. AOP Design basis will reflect the selected period as AOP basis, users can add additional basis points for summary.
- Constant tab is provided to enter the constant value which will not change month wise as per below snapshot.
- Report manual entry tab is provided to enter report specific manual entries for last budgeted FY actual values (e.g. Budgeted selectivity, Actual Selectivity, guaranteed selectivity and actual operating Hours)
- Option to import and export manually entered from configuration screen is provided.

The screenshot below highlights the details captured in the digitized AOP platform.

AOP Historical Period Basis

Start Date

01-02-2020

End Date

31-03-2025

Load

(Last refreshed data on: 01-07-2025 05:52 PM for the period from 01-02-2020 to 31-03-2025)

AOP Design Basis

1. AOP design basis is considered for the period from 01-02-2020 to 31-03-2025.

Configuration

Constants

Report Manual Entry (2024-25)

Download

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Save

Particu...	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
▼ Yields														
MEG Yield	%	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287 AOP FY25-26
PEG Yield	%	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24 AOP FY25-26
▼ Calculation input for Norm														
Predicted Select...	%	87.288	87.147	87.007	86.868	86.73	86.593	86.457	89.046	89.681	89.866	89.913	89.877	AOP FY25-26
TST	C	250.485	251.106	251.705	252.282	252.838	253.375	253.894	231.425	232.109	233.463	234.791	236.077	AOP FY25-26
Total HP Steam ...	TPH	0.486	0	0	0	3.495	6.25	1.398	3.889	0	0	0	0	0 AOP FY25-26
Total MP Steam ...	TPH	16.862	16.178	15.761	15.34	18.375	19.901	15.687	24.834	22.232	22.824	23.071	23.102	AOP FY25-26
Total LP Steam C...	TPH	5.082	5.082	5.082	5.147	5.082	5.082	5.082	6.501	5.082	5.082	5.153	5.082	AOP FY25-26
Ethylene purity		0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	12
Oxygen purity		0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	12
Ethylene losses		1.018	1.018	1.018	1.018	1.018	1.018	1.018	1.018	1.018	1.018	1.018	1.018	from dct update...
Oxygen losses		1.008	1.008	1.008	1.008	1.008	1.008	1.008	1.008	1.008	1.008	1.008	1.008	from dct update1
MP Latent Heat	kcal / kg	390.422	390.422	390.422	390.422	390.422	390.422	390.422	390.422	390.422	390.422	390.422	390.422	AOP FY25-26

Configuration

Constants

Report Manual Entry (2024-25)

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Save

Particulars	UOM	Value	Remark
▼ Constant			
BPW Feed Temperature	°C	120.0	From DCT
CO2 Molecular weight		44.02	From DCT
Cycle gas T	°C	35.0	From DCT
Cycle gas CP	kcal/kg/°C	0.5131	From DCT
Cycle gas flow	kg/h	210000.0	From DCT
EO molecular weight		44.05	From DCT
Ethylene Molecular Weight		28.05	From DCT
Oxygen molecular weight		32.0	From DCT
Stoichiometric Ratio		120.0	From DCT
Water Cp	kcal/kg/°C	1.05	From DCT

Configuration

Constants

Report Manual Entry (2024-25)

Download

Print

Save

Particulars	U...	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
▼ Report Manual Entry														
Budgeted Selectivity	%	87.288	87.147	87.007	86.868	86.73	86.593	86.457	89.046	89.681	89.866	89.913	86.4	AOP FY25-26
Actual Selectivity	%	87.288	87.147	87.007	86.868	86.73	86.593	86.457	89.046	89.681	89.866	89.913	86.4	AOP FY25-26
Guaranteed Selectivity	%	87.288	87.147	87.007	86.868	86.73	86.593	86.457	89.046	89.181	89.366	89.413	89.377	AOP FY25-26
Actual Operating Hours	Hrs	704	744	720	744	744	720	744	384	744	744	672	744	AOP updated

Figure 2 Digitized AOP screen for Configuration/manual Inputs

2.1.1.2 Production Volume Data

The Max Capacity (MC) of EOEG plants is defined on an EOE basis. MC values for Ethylene Oxide (EO), Ethylene Oxide Equivalent (EOE) and Monoethylene Glycol (MEG) will be captured separately.

Reliance Industries Ltd
AOP Automation for NMD EOEG Plant

Proprietary and Confidential
Document No.
November 5, 2025

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Month wise data is considered for deriving the best value. This will account for seasonal effects as well as the festive season.

If AOP basis is considered as 05 years, value estimation involves averaging production data over continuous 5-day periods in the last 5 years, after removing outliers and considering the normal operating mode without recycle for each month, 5 such averages are calculated and the maximum average value from the available 05 average set is designated as the MC value or production volume. Also, the edit option provided to change the derived MC value.

Below tags are considered for NMD site to identify normal operating mode without recycle.

- NEOEG_IN004
- NEOEG_IN008
- NEOEG_OUT004

NOTE- AOP historical period basis added in configuration screen where user can change Start date and End date to change the AOP design basis, accordingly MCU and Norms historical data will be considered for calculation.

A similar approach is employed for calculating the EO MC value.

The following logic has been formulated to derive month-wise EOE and EO MC values.

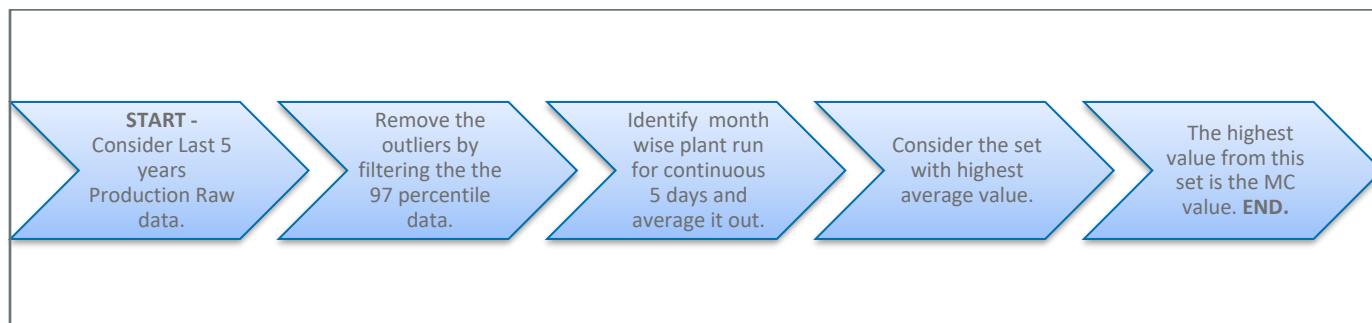


Figure 3 Digitized AOP Steps to derive EOE and EO month wise MC values

The MEG MC value is determined using the following formula.

$$MEG MC = MEG Yield * (EOE MC - EO MC)$$

Where MEG yield is a month-wise manual entry captured in the AOP digitization portal in the Configuration screen, as seen in the screenshot below.

Particu...	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Yields														
MEG Yield	%		1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287 / AOP FY25-26
PEG Yield	%		0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24 / AOP FY25-26

Figure 4 Digitized AOP screen for MEG Yield

If the user needs to modify the production rate this can be done by providing a specific reason.

The AOP digitization portal captures a month-wise percentage summary, comparing each month's value against the highest MU value recorded over the past 12 months.

This information is captured in the AOP digitization portal as seen in the screenshot below.



Production Volume Data

Year: 2025-26 Vertical: MEG Site: NMD Plant: EOEg

1 Configuration 2 Shutdown/TA Activities 3 Slowdown Activities **4 Production Volume Data** 5 Business Demand 6 Maintenance Details 7 Production AOP 8 Normal Operation Norms 9 Shutdown Norms 10 Slowdown Norms 11 Consumption AOP 12 Feed Stock Availability

MEG / NMD / EOEg




Save Calculate

Select UOM
TPH

Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
EOE	11.932	12	12	12	11.161	10.7	11.613	10.763	12	12	12	12	12 / Aop 25-26
EO	9.601	9.7	9.7	9.7	8.358	8.233	9.042	7.938	9.7	9.7	9.7	9.7	9.7 / Aop 25-26
MEG	3	2.961	2.961	2.961	3.609	3.175	3.31	3.637	2.961	2.961	2.961	2.961	2.961 / Aop 25-26

Percentage Summary

Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	
EOE	99.43	100	100	100	93.01	89.17	96.77	89.69	100	100	100	100	
EO	98.98	100	100	100	86.17	84.88	93.22	81.83	100	100	100	100	
MEG	82.5	81.42	81.42	81.42	99.23	87.32	91.01	100	81.42	81.42	81.42	81.42	

Figure 5 Digitized AOP screen for Production Volume data

The '**Calculate**' button on this screen allows the user to compute the MEG MC value if any changes have been made to the MEG Yield and Start and End date for AOP basis selection on the configuration screen.

2.1.1.3 Business Demand

The production demand is considered as the maximum value from last year's achieved production values. The default value for EOE, EO and MEG is 100%.

The business demand value can be modified by users upon providing a specific reason.

This information is captured in the AOP digitization portal as seen in the screenshot below.

NOTE- For the NMD EOEG plant, CO2 demand will be captured in this section.

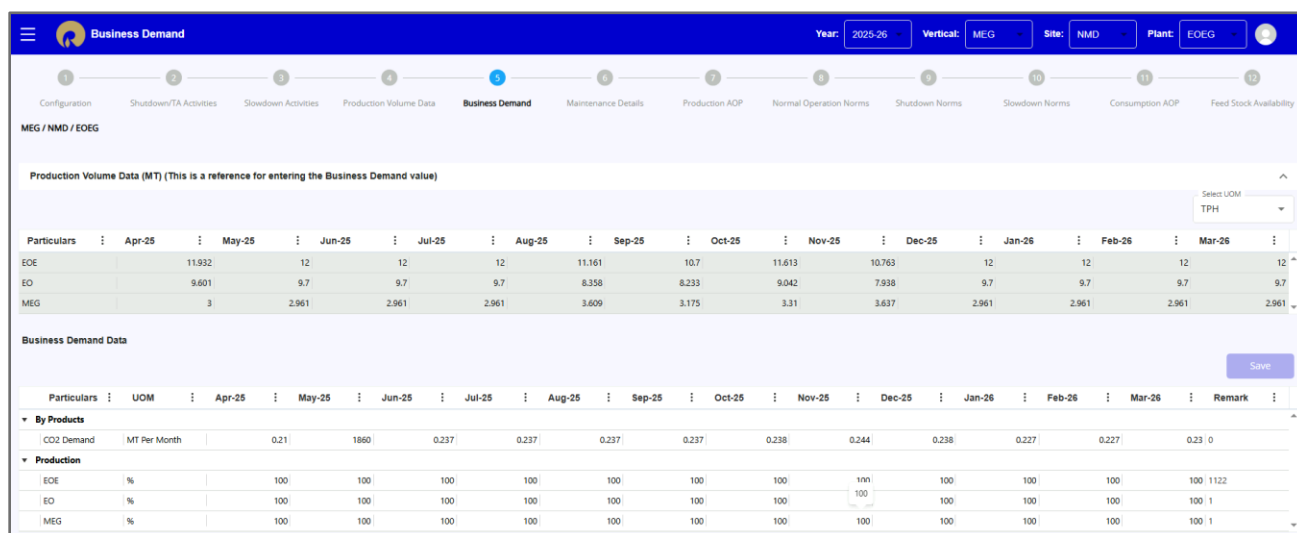


Figure 6 Digitized AOP screen for Business Demand

2.1.1.4 Maintenance/Shutdown/Slowdown/TA Activities

The operations team will provide the shutdown, turnaround and slowdown plan to calculate month-wise operating hours available for production.

The below screenshot depicts the shutdown and turnaround activities planned, as will be captured in the AOP digitized platform.

The user has to enter the SD-From and SD-To date and time to calculate the duration in hours, or SD-From and Duration(hrs) to calculate the SD-To date/time.

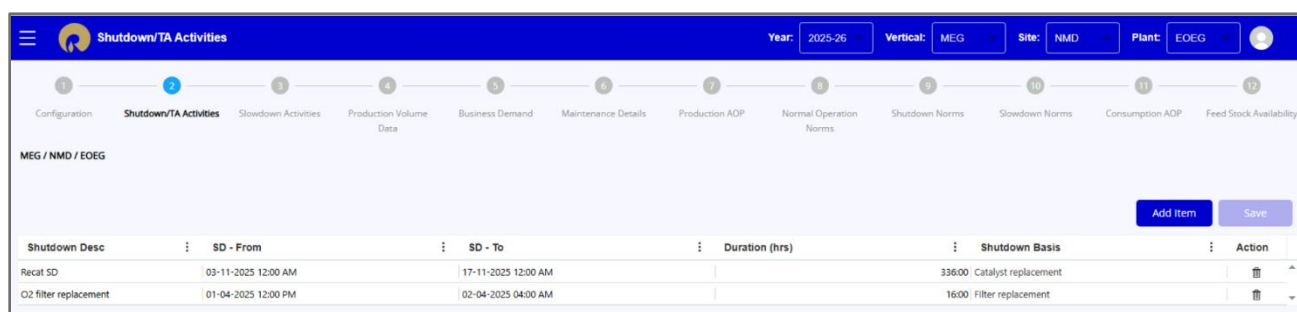


Figure 7 Digitized AOP screen for Shutdown/Turnaround Activities

The Slowdown entries will have the inputs as shutdowns, along with the requirement of plant EOE/EO load (TPH basis) for the duration of the slowdown.

The slowdown activities can be calculated based on the slowdown rate and hours. This is called a Basis Shutdown, equivalent to Shutdown.

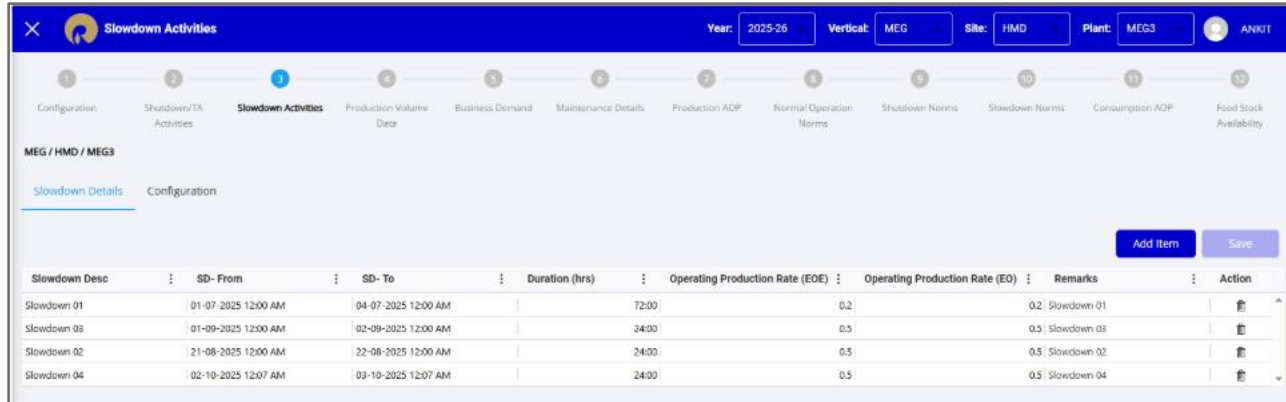
$$\text{Slowdown rate(TPH)} = (\text{MCU} - \text{Operating Production rate})$$

$$\text{Basis shutdown equivalent of Slowdown hrs} = (\text{Slowdown rate(TPH)} * \text{Slowdown hrs}) / \text{MCU}$$

Operating Production rate will be a manual entry in Slowdown activity screen and slowdown rate will be calculate at the backend as per above formula.

For each shutdown entry, two slowdown entries will be generated automatically, with the nomenclature as “#Shutdown Entry name. – Ramp down” and “#Shutdown Entry name. – Ramp Up”.

In Slowdown details tab, operating production rate-EOE (TPH) and operating production rate-EO (TPH) these two columns will be added in Slowdown activities screen as shown below.



Slowdown Desc	SD- From	SD- To	Duration (hrs)	Operating Production Rate (EOE)	Operating Production Rate (EO)	Remarks	Action
Slowdown 01	01-07-2025 12:00 AM	04-07-2025 12:00 AM	72:00	0.2		0.2 Slowdown 01	 
Slowdown 03	01-09-2025 12:00 AM	02-09-2025 12:00 AM	24:00	0.5		0.5 Slowdown 03	 
Slowdown 02	21-08-2025 12:00 AM	22-08-2025 12:00 AM	24:00	0.5		0.5 Slowdown 02	 
Slowdown 04	02-10-2025 12:07 AM	03-10-2025 12:07 AM	24:00	0.5		0.5 Slowdown 04	 

Separate tab (Configuration) will be provided as shown below in slowdown activity screen for the provision of manual entry related to configuration screen parameters based on the type of slowdown. Number of rows mentioned in slowdown activity screen will be no of column for slow down manual entry screen.

Particulars	Slowdown 01 (July)	Slowdown 02 (August)	Slowdown 03 (September)	Slowdown 04 (October)
MEG Yield	0.28	0.3	0.28	0.3
DEG Yield	0.11	0.12	0.18	0.3
TEG Yield	0.0063	0.0065	0.28	0.3
TST	0.23	0.25	0.28	0.3
Total HP Steam Consumption	0.34	0.35	0.28	0.3
Total MP Steam Consumption	0.3	0.31	0.28	0.3
Total LP Steam Consumption	0.09	0.1	0.28	0.3
Ethylene purity	0.099	0.099	0.28	0.3
Oxygen purity	0.097	0.097	0.28	0.3
Ethylene losses	0.101	0.101	0.28	0.3
Oxygen losses	0.101	0.055	0.28	0.3
HP Latent Heat	0.4	0.42	0.28	0.3

Figure 8 Digitized AOP screen for Slowdown Activities

The 'Add Item' button on the Shutdown/Slowdown activities screen allows users to insert a new entry for shutdown/slowdown.

The 'Save' button enables users to save newly added items as well as any modifications made to existing entries.

For each month, the following table is generated based on all inputs for shutdowns/slowdowns-

Description	April	... March
Total Available Hours for the Month	30*24=720	
Shutdowns Hours	From the Routine Shutdown Plan Section	
Non-Shutdown Operating Hours for the Month	Total Available operating hours for the month – Shutdown hrs	
Average Slowdown Load/rate for EOE, % of Production Volume Target (PVT)	$\frac{\sum_{i=1}^n (EOE \text{ slowdown hrs}_i \times EOE \text{ load/rate during slowdown}_i)}{\sum_{i=1}^n (EOE \text{ slowdown hrs}_i) \times EOE \text{ MC for Month}}$	
Average Slowdown Load/rate for EO, % of Production Volume Target (PVT)	$\frac{\sum_{i=1}^n (EO \text{ slowdown hrs}_i \times EO \text{ load/rate during slowdown}_i)}{\sum_{i=1}^n (EO \text{ slowdown hrs}_i) \times EO \text{ MC for Month}}$	

EOE Slowdown
Load/rate Reduction
as Shutdown Hours
Equivalent

$$EOE \text{ Average slowdown load/rate, \% of PVT} \times \sum_{i=1}^n (EOE \text{ slowdown hrs}_i)$$

EO Slowdown
Load/rate Reduction
as Shutdown Hours
Equivalent

$$EO \text{ Average slowdown load/rate, \% of PVT} \times \sum_{i=1}^n (EO \text{ slowdown hrs}_i)$$

EOE Effective
Operating Hours @
PVT

$$\begin{aligned} & \text{Total Available operating hours for the month} - \text{Shutdown/TA hours} \\ & - \text{EOE Slowdown load reduction as Shutdown hours Equivalent (other reasons)} \\ & - \text{EOE Slowdown load reduction as Shutdown hours Equivalent (Rampup/RampDown)} \end{aligned}$$

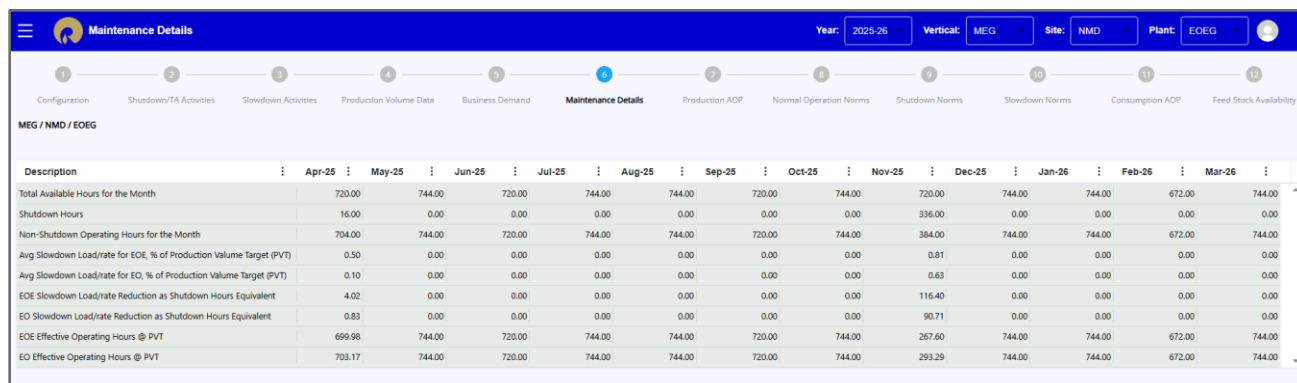
EO Effective Operating
Hours @ PVT

$$\begin{aligned} & \text{Total Available operating hours for the month} - \text{Shutdown/TA hours} \\ & - \text{EO Slowdown load reduction as Shutdown hours Equivalent (other reasons)} \\ & - \text{EO Slowdown load reduction as Shutdown hours Equivalent (Rampup/RampDown)} \end{aligned}$$

Table 3 Maintenance Details

The screen below depicts the Maintenance Details Table, which is captured in the digitized platform.

NOTE- The below screen is read-only. The users cannot change the values.



	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
Total Available Hours for the Month	720.00	744.00	720.00	744.00	744.00	720.00	744.00	720.00	744.00	744.00	672.00	744.00
Shutdown Hours	16.00	0.00	0.00	0.00	0.00	0.00	0.00	336.00	0.00	0.00	0.00	0.00
Non-Shutdown Operating Hours for the Month	704.00	744.00	720.00	744.00	744.00	720.00	744.00	384.00	744.00	744.00	672.00	744.00
Avg Slowdown Load/rate for EOE, % of Production Volume Target (PVT)	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.81	0.00	0.00	0.00	0.00
Avg Slowdown Load/rate for EO, % of Production Volume Target (PVT)	0.10	0.00	0.00	0.00	0.00	0.00	0.00	0.63	0.00	0.00	0.00	0.00
EOE Slowdown Load/rate Reduction as Shutdown Hours Equivalent	4.02	0.00	0.00	0.00	0.00	0.00	0.00	116.40	0.00	0.00	0.00	0.00
EO Slowdown Load/rate Reduction as Shutdown Hours Equivalent	0.83	0.00	0.00	0.00	0.00	0.00	0.00	90.71	0.00	0.00	0.00	0.00
EOE Effective Operating Hours @ PVT	699.98	744.00	720.00	744.00	744.00	720.00	744.00	267.60	744.00	744.00	672.00	744.00
EO Effective Operating Hours @ PVT	703.17	744.00	720.00	744.00	744.00	720.00	744.00	293.29	744.00	744.00	672.00	744.00

Figure 9 AOP Maintenance Details Table

2.1.1.5 Calendar Year Based Production Calculations Details/Production AOP

Using the Business Demand, Production Volume data and Maintenance Plan (Operating Hours), the month-wise production quantity is derived.

The month-wise budget production for each product is calculated in MT and KT, the formula for which is mentioned below.

$$\begin{aligned} \text{Production AOP (MT)} &= (\text{Operating hours for EOE/EO in a month (hrs)} * \\ & \text{Percentage of product demand} * \text{MC Value of product(TPH)}) / 100 \end{aligned}$$

$$\text{Operating hours for MEG} = \text{Total Available hours of the month} - \text{Shutdown/TA hours of the month}$$

Operating hours for EOE = Total Available operating hours for the month – Shutdown/TA hours – EOE Slowdown load reduction as Shutdown hours Equivalent (other reasons) – EOE Slowdown load reduction as Shutdown hours Equivalent (Ramp up/Ramp Down)

Operating hours for EO = Total Available operating hours for the month – Shutdown/TA hours – EO Slowdown load reduction as Shutdown hours Equivalent (other reasons) – EO Slowdown load reduction as Shutdown hours Equivalent (Ramp up/Ramp Down)

Hence, month-wise production is,

$$\text{Total Budgeted Production} = \sum \text{All the values for the month}$$

The month-wise budget production is further used to derive the annual plan in KT, and this is part of the final report.

This information is captured in the AOP digitization portal as seen in the screenshot below.

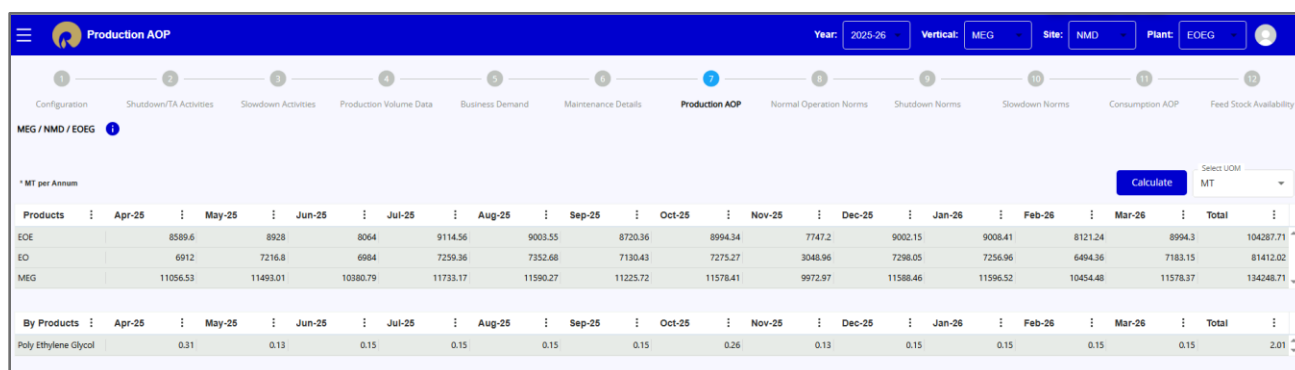


Figure 10 Digitized AOP screen for Production AOP month wise

The '**Calculate**' button on this screen allows users to recalculate Production values when changes are made to configuration settings (e.g., MEG yield), business demand, or when line items are added to slowdown and shutdown activities.

2.1.2 Norms

Below are the steps captured to derive the Norms value.

2.1.2.1 Normal Operation Norms

All the specific consumption norms are set with One-unit EOE as the basis.

For calculation purpose in the Digitized AOP platform, based on the DCT provided by the user, Norms are broadly classified as –

1. Historical Best
2. Based on Calculations
3. Manual Entries

3.1.2.1.1 Historical Best

Based on the historical data of last 5 years, corresponding to the MCU production, the best-achieved feed slate period is filtered. This data is further used for the calculation of consumption norms, which is the average of the norms during that period, excluding the identified outliers.

Below is the logic formulated to derive the Historical best Normal Operation Norms –

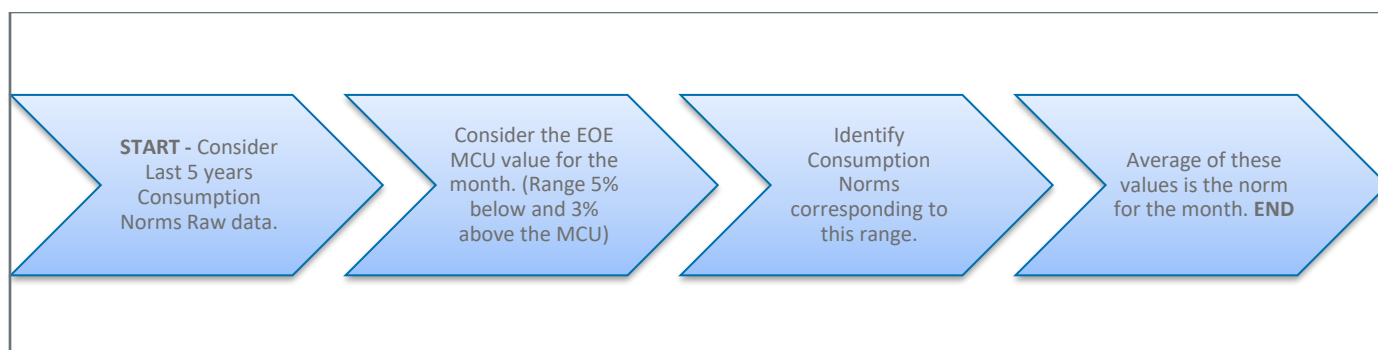


Figure 11 Digitized AOP Steps to derive month wise Historical Best Norms

The user has provision to overwrite/update the value by providing a valid reason for the historical best particulars.

The list of particulars to be derived using the Historical Best logic is specified in the DCT provided by the user. These particulars may vary from plant to plant.

3.1.2.1.2 Based on Calculations

For the MEG vertical, particulars like ethylene, oxygen, HP Steam, BFW water etc. are formulated based on the stoichiometric calculations provided by the user in the DCT.

The formula can vary from plant to plant.

E.g. the Raw Material norms for ethylene and oxygen for NMD EOEG are derived using the calculations mentioned in the below table.

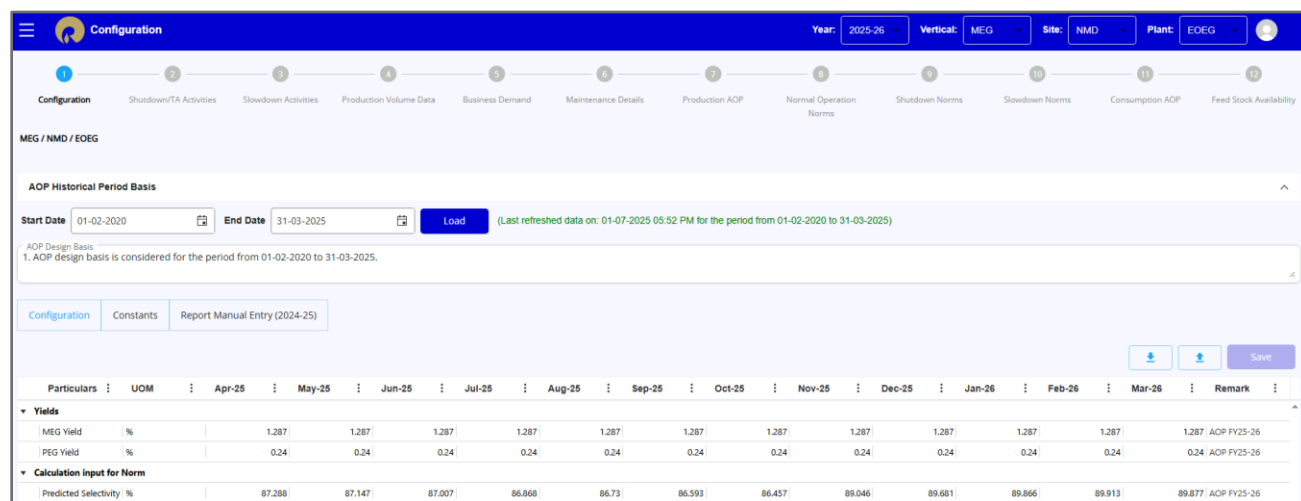
Plant	Category	Particular	Calculation
NMD EOEG	Raw Material	Ethylene	<i>Ethylene Mol Wt/EO Mol Wt*100/{selectivity}*Ethylene Losses/Ethylene Purity</i> <i>Where,</i> <i>Ethylene Molecular Weight - 28.05</i> <i>EO molecular weight - 44.05</i> <i>Ethylene losses - 1.018</i> <i>Ethylene Purity (wt%) - 0.997</i>

Plant	Category	Particular	Calculation
		Oxygen	$\text{Oxygen Mol Wt/EO Mol Wt} \times \left(\frac{120 - \{\text{selectivity}\}}{\{\text{selectivity}\}} \right) \times 40 \times \text{Oxygen Losses/Oxygen Purity}$ Where, Oxygen molecular weight - 32 EO molecular weight - 44.05 Stoichiometric Ratios - 120,40 Oxygen losses- 1.008 Oxygen Purity - 0.997

Table 4 Raw Material Norms calculations

The values for *losses and purity* can be captured from the Lab analysis or Past/last year's operations. The user has the provision to enter the value manually in the Configuration screen.

Selectivity, Losses and Purity can be modified/added by the user in the Configuration Screen under the Calculation Input for Norms Subsection.



Configuration Year: 2025-26 Vertical: MEG Site: NMD Plant: EDEG

1 Configuration 2 Shutdown/TA Activities 3 Slowdown Activities 4 Production Volume Data 5 Business Demand 6 Maintenance Details 7 Production AOP 8 Normal Operation Norms 9 Shutdown Norms 10 Slowdown Norms 11 Consumption AOP 12 Feed Stock Availability

MEG / NMD / EOEG

AOP Historical Period Basis

Start Date: 01-02-2020 End Date: 31-03-2025 Load (Last refreshed data on: 01-07-2025 05:52 PM for the period from 01-02-2020 to 31-03-2025)

AOP Design Basis
1. AOP design basis is considered for the period from 01-02-2020 to 31-03-2025.

Configuration Constants Report Manual Entry (2024-25)

Particulars : UOM : Apr-25 : May-25 : Jun-25 : Jul-25 : Aug-25 : Sep-25 : Oct-25 : Nov-25 : Dec-25 : Jan-26 : Feb-26 : Mar-26 : Remark :

Yields

MEG Yield	%	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	AOP FY25-26
PEG Yield	%	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	AOP FY25-26

Calculation input for Norm

Predicted Selectivity	%	87.288	87.147	87.007	86.868	86.73	86.593	86.457	86.321	86.185	86.049	85.913	85.777	85.641	AOP FY25-26
-----------------------	---	--------	--------	--------	--------	-------	--------	--------	--------	--------	--------	--------	--------	--------	-------------

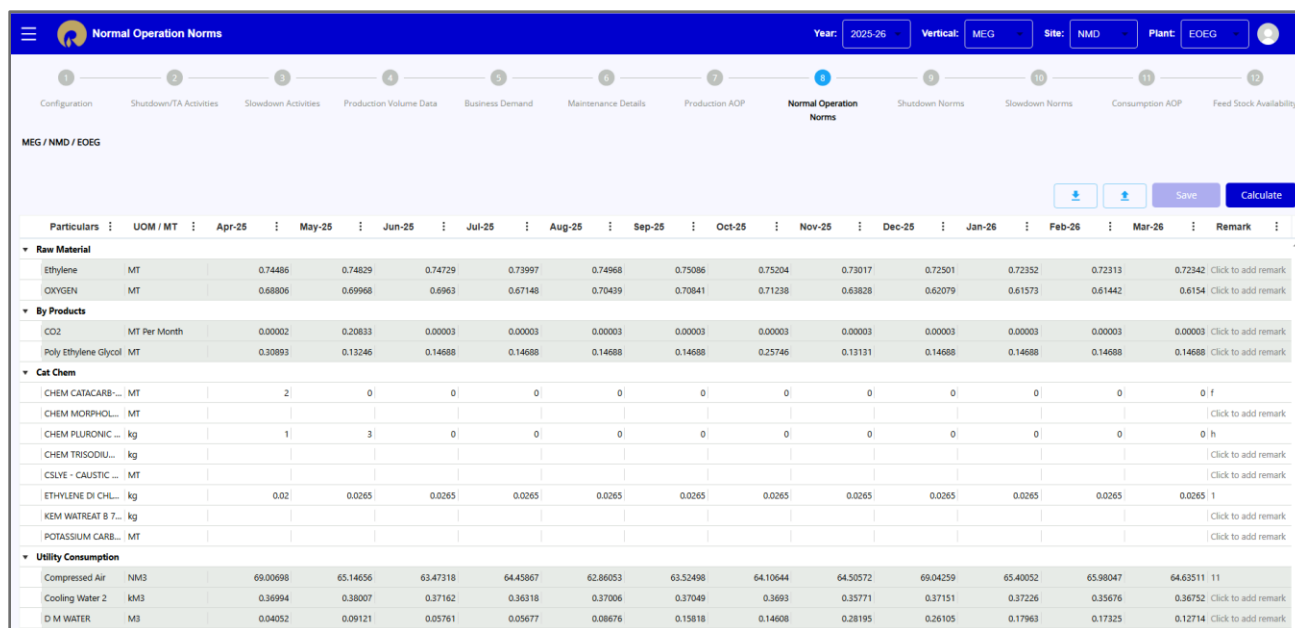
Figure 12 Digitized AOP screen for Catalyst Selectivity month wise

In the Normal Operations Norms screen the cells for particulars with formula, e.g. HP Steam Dis, will be read only. If the values for such particulars are to be modified then the respective inputs should be modified, e.g. Total HP Steam consumption in the Configuration Screen.

3.1.2.1.3 Manual Entries

The particulars under the Chemicals and Catalyst sections are manually entered by the user, who must also provide the corresponding norm values in the Normal Operation Norms screen.

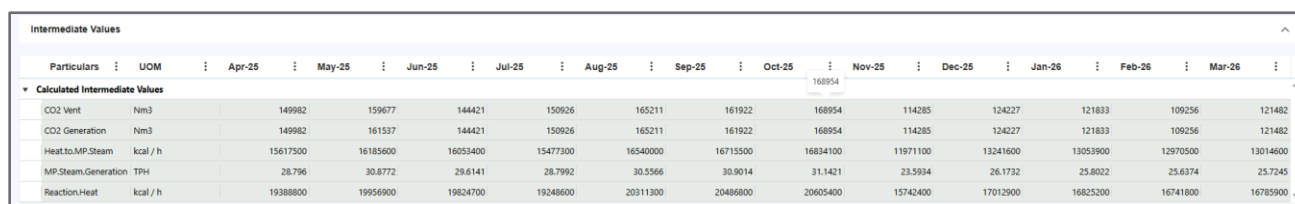
The screenshot below highlights the details captured in the digitized AOP platform in the Normal Operations Norms screen.



Particulars	UOM / MT	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Raw Material														
Ethylene	MT	0.74486	0.74829	0.74729	0.73997	0.74968	0.75086	0.75204	0.73017	0.72501	0.72352	0.72313	0.72342	Click to add remark
OXYGEN	MT	0.68806	0.69968	0.6963	0.67148	0.70439	0.70841	0.71238	0.63828	0.62079	0.61573	0.61442	0.6154	Click to add remark
By Products														
CO2	MT Per Month	0.00002	0.20833	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003	Click to add remark
Poly Ethylene Glycol	MT	0.30893	0.13246	0.14688	0.14688	0.14688	0.14688	0.25746	0.13131	0.14688	0.14688	0.14688	0.14688	Click to add remark
Cat Chem														
CHEM CATAcarb...	MT	2	0	0	0	0	0	0	0	0	0	0	0	Click to add remark
CHEM MORPHOL...	MT													Click to add remark
CHEM PLURONIC ...	kg	1	3	0	0	0	0	0	0	0	0	0	0	Click to add remark
CHEM TRISODIUM...	kg													Click to add remark
CSLYE - CAUSTIC ...	MT													Click to add remark
ETHYLENE DI CHL...	kg	0.02	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	Click to add remark
KEM WATREAT B 7...	kg													Click to add remark
POTASSIUM CARB...	MT													Click to add remark
Utility Consumption														
Compressed Air	NM3	69.00698	65.14656	63.47318	64.45867	62.86053	63.52498	64.10644	64.50572	69.04259	65.40052	65.98047	64.63511	11
Cooling Water 2	km3	0.36994	0.38007	0.37162	0.36318	0.37006	0.37049	0.3693	0.35771	0.37151	0.37226	0.35676	0.36752	Click to add remark
D M WATER	M3	0.04052	0.09121	0.05761	0.05677	0.08676	0.15818	0.14608	0.28195	0.26105	0.17963	0.17325	0.12714	Click to add remark

Figure 13 Digitized AOP screen for Normal operations Norms

The Normal Operation Normas screen also captures the Intermediate calculations required for Norms calculation as shown in the below screen shot –



Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
Calculated Intermediate Values													
CO2 Vent	Nm3	149982	159677	144421	150926	165211	161922	168954	114285	124227	121833	109256	121482
CO2 Generation	Nm3	149982	161537	144421	150926	165211	161922	168954	114285	124227	121833	109256	121482
Heat.to.MP.Steam	kcal / h	15617500	16185600	16053400	15477300	16540000	16715500	16834100	11971100	13241600	13053900	12970500	13014600
MP.Steam.Generation	TPH	28.796	30.8772	29.6141	28.7992	30.5566	30.9014	31.1421	23.5934	26.1732	25.8022	25.6374	25.7245
Reaction.Heat	kcal / h	19388800	19956900	19824700	19248600	20311300	20486800	20605400	15742400	17012900	16825200	16741800	16785900

Figure 14 Digitized AOP screen for Normal operations Norms (Calculated Intermediated values)

The **'Calculate'** button on the screen will enable if the user makes any change in the dependent inputs in the Configuration screen.

The **'Save'** button on the above screen enables the user to save any norm value updated manually for historical best norms for cat chem.

2.1.2.2 Shutdown Norms

The Shutdown norms are recorded for the months in which shutdown activities are planned, as specified in the Shutdown Activities screen.

The shutdown quantity utilization data includes TA, asset maintenance linked shutdown/ slowdown activities, etc..

The use must enter the consumption data (absolute values) for each line item.

The below screenshot depicts the Shutdown Norms captured in the AOP digitized platform.

Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Raw Material														
Ethylene	MT													Click to add remark
OXYGEN	MT													Click to add remark
By Products														
CO2	MT Per Month													Click to add remark
Poly Ethylene Glycol	MT													Click to add remark
Cat Chem														
CHEM CATAB... MT														Click to add remark
CHEM MORPHOL... MT														Click to add remark
CHEM PLURONIC ... kg														Click to add remark
CHEM TRISODIU... kg														Click to add remark
CSLYE - CAUSTIC ... MT														Click to add remark
ETHYLENE DI CHL... kg														Click to add remark
KEM WATREAT 8 7... kg														Click to add remark
POTASSIUM CARB... MT														Click to add remark
Utility Consumption														
Compressed Air	NM3													Click to add remark
Cooling Water 2	KM3													Click to add remark
D M WATER	M3													Click to add remark

Figure 14 Digitized AOP screen for Shutdown Norms

2.1.2.3 Slowdown Norms

Slowdown norms are recorded for the months in which slowdown activities are planned, as specified in the Slowdown Activities screen.

The values for slowdown norms are same as the Normal Operations norms for the month.

NOTE- Slowdown wise norms calculation will be provided by RIL.

The below screenshot depicts the Shutdown Norms captured in the AOP digitized platform.

Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Raw Material														
Ethylene	MT	0.37243	0	0	0	0	0	0	0	0.36509	0	0	0	Updated by system
OXYGEN	MT	0.34403	0	0	0	0	0	0	0	0.31914	0	0	0	Updated by system
By Products														
CO2	MT Per Month	0.00001	0	0	0	0	0	0	0	0.00002	0	0	0	Updated by system
Poly Ethylene Glycol	MT	0.15446	0	0	0	0	0	0	0	0.06565	0	0	0	Updated by system
Cat Chem														
CHEM CATACARB...	MT	1	0	0	0	0	0	0	0	0	0	0	0	Updated by system
CHEM MORPHOL...	MT	0	0	0	0	0	0	0	0	0	0	0	0	Updated by system
CHEM PLURONIC ...	kg	0.5	0	0	0	0	0	0	0	0	0	0	0	Updated by system
CHEM TRISODIU...	kg	0	0	0	0	0	0	0	0	0	0	0	0	Updated by system
CSLYE - CAUSTIC ...	MT	0	0	0	0	0	0	0	0	0	0	0	0	Updated by system
ETHYLENE DI CHL...	kg	0.01	0	0	0	0	0	0	0	0.01234	0	0	0	Updated by system
KEM WATREAT B 7...	kg	0	0	0	0	0	0	0	0	0	0	0	0	Updated by system
POTASSIUM CARB...	MT	0	0	0	0	0	0	0	0	0	0	0	0	Updated by system
Utility Consumption														
Compressed Air	NM3	34.50349	0	0	0	0	0	0	0	32.25286	0	0	0	Updated by system
Cooling Water 2	km3	0.18497	0	0	0	0	0	0	0	0.17886	0	0	0	Updated by system
D M WATER	M3	0.02026	0	0	0	0	0	0	0	0.14098	0	0	0	Updated by system

Figure 15 Digitized AOP screen for Shutdown Norms

2.1.2.4 Budget Norms/Consumption AOP

Using the values of Normal operations Norms, Shutdown Consumptions and Slowdown Norms, the final Consumption AOP value is calculated as below –

Consumption AOP for months without shutdown or slowdown = Normal Operations Norms

Consumption AOP for months with shutdown or slowdown = Total Consumption / Total production

Where,

*Total Consumption = (Normal Operation Norms * EOE MC value * EOE operating hours) + $\sum_{i=1}^n$ (Slowdown Norms * EOE slowdown hours * EOE slowdown rate)_n + Shutdown Consumption*

*Total Production = (EOE MC value * EOE operating hours) + $\sum_{i=1}^n$ (EOE slowdown hours * EOE slowdown rate)_n*

The below screenshot depicts the Shutdown Norms captured in the AOP digitized platform.

Consumption AOP

Year:2025-26Vertical:MEGSite:NMDPlant:EDEG

1

2

3

4

5

6

7

8

9

10

11

12

ConfigurationShutdown/TA ActivitiesSlowdown ActivitiesProduction Volume DataBusiness DemandMaintenance DetailsProduction AOPNormal Operation NormsShutdown NormsSlowdown NormsConsumption AOPFeed Stock Availability

MEG / NMD / EOEG 1

Calculate

Particulars	UOM / MT	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	
Raw Material														
Ethylene	MT		0.74486	0.74829	0.74729	0.73997	0.74968	0.75086	0.75204	0.73017	0.72501	0.72352	0.72313	0.72342
OXYGEN	MT		0.68806	0.69968	0.6963	0.67148	0.70439	0.70841	0.71238	0.63828	0.62079	0.61573	0.61442	0.6154
By Products														
CO2	MT Per Month		0.00002	0.20833	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003
Poly Ethylene Glycol	MT		0.30893	0.13246	0.14688	0.14688	0.14688	0.14688	0.25746	0.13131	0.14688	0.14688	0.14688	0.14688
Cat Chem														
CHEM CATAcarb-9...	MT													
CHEM MORPHOLENE	MT													
CHEM PLURONIC P...	kg		1	3	0	0	0	0	0	0	0	0	0	0
CHEM TRISODIUM ...	kg													
CSLYE - CAUSTIC S...	MT								0.0265					
ETHYLENE DI CHLO...	kg		0.02	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265
KEM WATREAT B 70M	kg													
POTASSIUM CARBO...	MT													
Utility Consumption														
Compressed Air	NM3		65	65.14656	63.40943	64.34499	62.86053	63.52498	64.10644	64.50572	69.04259	65.40052	65.51126	63.65744
Cooling Water 2	km3		0.36994	0.38007	0.37134	0.36274	0.37006	0.37049	0.3693	0.35771	0.37151	0.37226	0.36362	0.37608
D M WATER	M3		0.04052	0.09121	0.05617	0.05648	0.08676	0.15818	0.14608	0.28195	0.26105	0.17963	0.21217	0.13624

Figure 16 Digitized AOP screen for Budget Norms or Consumption Norms

2.1.3 Feedstock Availability

Availability of Feedstock/Raw Material defines maximum attainable production in a plant. In the case of MEG technology, the availability of Ethylene and Oxygen will be accounted for planning of unit production.

For the Rev 0 for AOP production/norms will consider the 100% availability of the required.

- These quantities will be compared against feedstock requirements and used for optimizing the production to meet the constrained feedstock availability as per below mentioned logic.

Reduced No of Operating Hrs to Maximize the production by making Delta Feedstock (difference of Available and Required feedstock) = 0

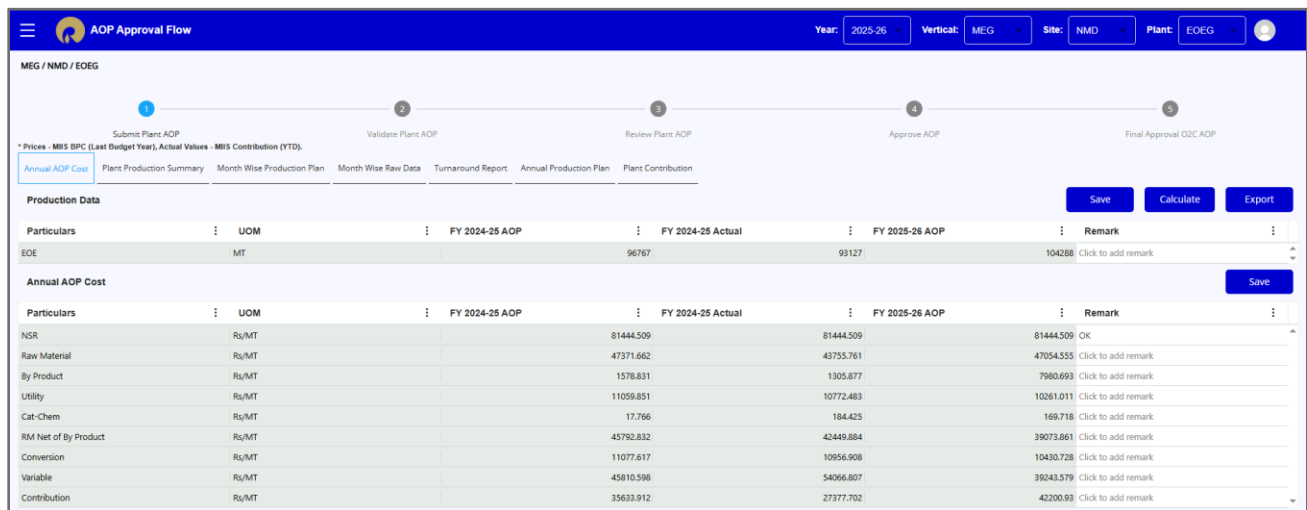
2.1.4 Other Cost (To generate cost sheet - Business)

- Process consumables will be discussed with RIL and TBD for a path forward.
- Maintenance consumables will be discussed with RIL and TBD for a path forward.

3. AOP Reports

- Dataset (Basis)
- Annual AOP Cost.
- Plant Production Summary (T-14).
- Month wise Production Plan (T – 16)
- Month wise Raw data (T – 18).
- Turnaround Report (T – 19A).
- Annual Production Plan (T – 15).
- Plant Contribution (T – 21).

The below screenshot depicts the AOP report captured in the AOP digitized platform.



The screenshot shows the 'AOP Approval Flow' interface. At the top, there are filters for Year (2025-26), Vertical (MEG), Site (NMD), and Plant (EOEG). Below these, a progress bar indicates five steps: 1. Submit Plant AOP, 2. Validate Plant AOP, 3. Review Plant AOP, 4. Approve AOP, and 5. Final Approval O2C AOP. The 'Annual AOP Cost' tab is selected, showing a table with columns: Particulars, UOM, FY 2024-25 AOP, FY 2024-25 Actual, FY 2025-26 AOP, and Remark. The table lists various cost components including NSR, Raw Material, By Product, Utility, Cat-Chem, RM Net of By Product, Conversion, Variable, and Contribution, with their respective values and remarks.

Particulars	UOM	FY 2024-25 AOP	FY 2024-25 Actual	FY 2025-26 AOP	Remark
EOE	MT		96767	93127	104288 Click to add remark
Annual AOP Cost					
NSR	Rs/MT		81444.509	81444.509	81444.509 OK
Raw Material	Rs/MT		47371.662	43755.761	47054.555 Click to add remark
By Product	Rs/MT		1578.831	1305.677	7980.693 Click to add remark
Utility	Rs/MT		11059.851	10772.483	10261.011 Click to add remark
Cat-Chem	Rs/MT		17.766	184.425	169.718 Click to add remark
RM Net of By Product	Rs/MT		45792.832	42449.884	39073.861 Click to add remark
Conversion	Rs/MT		11077.617	10956.908	10430.728 Click to add remark
Variable	Rs/MT		45810.598	54066.807	39243.579 Click to add remark
Contribution	Rs/MT		35633.912	27377.702	42200.93 Click to add remark

Figure 17 Digitized AOP screen for Annual AOP Cost

AOP Approval Flow

Year:

2025-26

Vertical:

MEG

Site:

NMD

Plant:

EOEG

MEG / NMD / EOEG

1

2

3

4

5

Submit Plant AOP

Validate Plant AOP

Review Plant AOP

Approve AOP

Final Approval O2C AOP

Prices - MIS BPC (Last Budget Year), Actual Values - MIS Contribution (YTD).

Annual AOP Cost

Plant Production Summary

Month Wise Production Plan

Month Wise Raw Data

Turnaround Report

Annual Production Plan

Plant Contribution

Plant Production Summary (T-14)

Save

	Item		2024-25		2025-26		Variance wrt current year budget		Variance wrt current year actuals		
SL.No	Production Volume	Unit	Budget	Actual	Budget	MT	%	MT	%	Remarks	
1	EOE	MT	96767.131	93127.305	104287.709	7520.577	7.772	11160.404	11.984	Click to add remark	
2	MEG	MT	27549	26355.036	134248.708	106699.708	387.309	107893.672	409.385	Click to add remark	
3	EO	MT	74907	72825.13	81412.018	6505.018	8.684	8586.888	11.791	Click to add remark	
4	Production	MT	96767.131	93127.305	104287.709	7520.577	7.772	11160.404	11.984	Click to add remark	
5	Operating Hours	Hrs	8760	8760		0	0	0	0	Click to add remark	
6	ProductionRate	TPH	11.046	10.631	11.969	0.922	8.349	1.338	12.584	Click to add remark	
7	NSR	Ru/MT	81444.509	81444.509	81444.509	0	0	0	0	Click to add remark	
8	Raw Material	Ru/MT	47371.662	43755.761	47054.555	317.108	0.669	-3298.793	-7.539	Click to add remark	
9	By Product	Ru/MT	1578.831	1305.877	7980.693	6401.862	405.481	6674.816	511.137	Click to add remark	
10	RM Net of By Product	Ru/MT	43792.832	42449.884	39073.861	6718.97	14.673	3376.023	7.953	Click to add remark	
11	Cat-Chem	Ru/MT	17.766	184.425	169.718	-151.952	-855.3	14.707	7.974	Click to add remark	
12	Utility	Ru/MT	11059.851	10772.483	10261.011	798.84	7.223	511.472	4.748	Click to add remark	
13	Conversion	Ru/MT	11077.617	10956.908	10430.728	646.888	5.84	526.179	4.802	Click to add remark	
14	Variable	Ru/MT	45810.598	54066.807	39243.579	6567.018	14.335	14823.228	27.417	Click to add remark	
15	Contribution	Ru/MT	35633.912	27377.702	42200.93	6567.018	18.429	14823.228	54.143	Click to add remark	
16	Contribution Rs Lacs	Rs lacs	34481.914	25496.116	44010.383	9528.469	27.633	18514.267	72.616	Click to add remark	

Figure 18 Digitized AOP screen for Production Plant Summary

AOP Approval Flow

Year:2025-26

Vertical:MEG

Site:NMD

Plant:EOEG

MEG / NMD / EOEG

1

2

3

4

5

Submit Plant AOP

Validate Plant AOP

Review Plant AOP

Approve AOP

Final Approval O2C AOP

* Prices - MIS BPC (Last Budget Year), Actual Values - MIS Contribution (YTD).

Annual AOP Cost

Plant Production Summary

Month Wise Production Plan

Month Wise Raw Data

Turnaround Report

Annual Production Plan

Plant Contribution

Monthwise Production (T-16)

Sl....	Month	2024-25				2025-26				Remarks			
		EOE Production, MT		Operating Hours		Throughput, TPH		Operating ...					
		Budget	Actual	Budget	Actual	Budget	Actual	MEG Throu...	EO Throug...	EOE Throu...	Total EOE, ...		
1	April	2376	4278	720	720	3.3	5.94	720	3	9.6	11.93	8589.6	Click to add remark
2	May	8424	1857	744	744	11.32	2.5	744	2.96	9.7	12	8928	Click to add remark
3	June	8640	8657	720	720	12	12.02	720	2.96	9.7	12	8640	Click to add remark
4	July	8928	9001	744	744	12	12.1	744	3.21	9.76	12.25	9114.56	Click to add remark
5	August	8928	8979	744	744	12	12.07	744	2.86	9.88	12.1	9003.55	Click to add remark
6	September	7584	8059	720	720	10.53	11.19	720	2.84	9.9	12.11	8720.36	Click to add remark
7	October	8928	8819	744	744	12	11.85	744	2.97	9.78	12.09	8994.34	Click to add remark
8	November	8352	8487	720	720	11.6	11.79	720	3.63	7.94	10.76	7747.2	Click to add remark
9	December	8928	8980	744	744	12	12.07	744	2.95	9.81	12.1	9002.15	Click to add remark
10	January	8928	8995	744	744	12	12.09	744	3.03	9.75	12.11	9008.41	Click to add remark
11	February	7824	8084	672	672	11.64	12.03	672	3.12	9.66	12.09	8121.24	Click to add remark
12	March	8928	8930	744	744	12	12	744	3.13	9.65	12.09	8994.3	Click to add remark
13	Total	96768	93126	8760	8760	11.05	10.63	8760	3.05	9.6	11.97	104863.71	Click to add remark

Main Products - Production for the budget year

Products	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total
EOE	8589.6	8928	8064	9114.56	9003.55	8720.36	8994.34	7747.2	9002.15	9008.41	8121.24	8994.3	104287.71

Figure 19 Digitized AOP screen for Month Wise Production Plan

AOP Approval Flow

Year: 2025-26

Vertical: MEG

Site: NMD

Plant: EOE

MEG / NMD / EOE

1

2

3

4

5

Submit Plant AOP

Validate Plant AOP

Review Plant AOP

Approve AOP

Final Approval O2C AOP

* Prices - MIS BPC (Last Budget Year), Actual Values - MIS Contribution (YTD).

Annual AOP Cost

Plant Production Summary

Month Wise Production Plan

Month Wise Raw Data

Turnaround Report

Annual Production Plan

Plant Contribution

Monthwise Consumption (T-18)

Parameters	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	
Budgeted Selectivity L...	88.7	88.8	88.2	88	87.8	87.6	87.4	87.2	87	86.8	86.6	86.4	
Actual Selectivity for L...	89.22	89	88.77	88.68	88.49	88.34	88.16	87.93	87	86.8	86.6	86.4	
Guaranteed Selectivity...	86.2	85.9	85.6	85.3	85	84.7	84.3	84	86.9	87.2	87.4	87.6	
Predicted Selectivity L...	87.29	86.89	87.006	87.867	86.729	86.592	86.457	89.046	89.68	89.865	89.913	89.877	

RawMaterial

Particul...	UOM	Spec	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total	
OXYGEN	MT	0.665697	5910.143	6246.725	5614.947	6120.262	6341.972	6177.549	6407.379	4944.883	5588.415	5546.738	4989.854	5535.112	69423.98	
Ethylene	MT	0.738262	6398.049	6680.724	6026.163	6744.49	6749.761	6547.807	6764.076	5656.781	6526.637	6517.736	5872.723	6506.659	76991.607	

ByProducts

Particul...	UOM	Spec	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total	
Poly Ethylene Gly...	MT / MT EOE		2878.238	2046.485	1747.302	1771.029	1843.021	2011.703	2299.844	1966.408	2280.58	2533.685	1874.933	1867.4	25120.651	
Poly Ethylene Gly...	MT	0.167373	2653.559	1182.63	1184.448	1338.755	1322.45	1280.856	2315.6	1017.246	1322.244	1323.164	1192.856	1321.092	17454.984	
Carbon Dioxide	MT Per Month	0.0178598	0.21	1859.997	0.237	0.237	0.237	0.237	0.238	0.244	0.238	0.227	0.227	0.23	1862.559	

CatChem

Particul...	UOM	Spec	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total	
CAUSTIC SODA L...	MT															
CHEM CATAcarb...	MT															

Figure 20 Digitized AOP screen for Month Wise Raw Data

AOP Approval Flow

Year:

2025-26

Vertical:

MEG

Site:

NMD

Plant:

EOEG

MEG / NMD / EOEG

1

2

3

4

5

Submit Plant AOP

Validate Plant AOP

Review Plant AOP

Approve AOP

Final Approval O2C AOP

* Prices - MIS BPC (Last Budget Year), Actual Values - MIS Contribution (YTD).

Annual AOP Cost

Plant Production Summary

Month Wise Production Plan

Month Wise Raw Data

Turnaround Report

Annual Production Plan

Plant Contribution

Turnaround Details (T-19A)

SL.No

Activities

Turnaround Period

From

To

Duration, hrs

Remark

1

O2 filter

01-Apr-25

01-Apr-25

1600

Click to add remark

2

SD

01-Nov-25

15-Nov-25

33600

Click to add remark

Turnaround details for the previous years data since commissioning

SL.No

Activities

Turnaround Period From

Turnaround Period To

Duration, hrs

Period in Months

Remark

Action

1

O2 filter

01-Apr-25

01-Apr-25

16

Click to add remark

2

SD

01-Nov-25

15-Nov-25

336

Click to add remark

Figure 21 Digitized AOP screen for Turnaround Report

Figure 22 Digitized AOP screen for Annual Production Plan

Figure 23 Digitized AOP screen for Plant Contribution

4. Key Assumptions and Exemptions

The above scope of work should be read with the following assumptions made

1. Data Acceptance
2. DCT provided by NMD Site team

5. Appendix

5.1 A1 - Manual Entry Details

Following are the details of Manual entries to be updated on Digital AOP portal.

Screen Name	Category	Manual Entry / Particulars	UoM	Reason
Production and Norms Basis	Production Basis - Calculation	MEG Yield	%	No Calculation Available.
	Norms Basis - Calculation	PEG Yield	%	No Calculation Available.
		Predicted Selectivity	%	No Calculation Available.
		TST	C	No Temperature transmitter.
		Total HP Steam Consumption	TPH	No Calculation Available.
		Total MP Steam Consumption	TPH	No Calculation Available.
		Total LP Steam Consumption	TPH	No Calculation Available.
		Ethylene Purity	wt%	Constant.
		Oxygen Purity	wt%	Constant.
		Ethylene Losses		Constant.
		Oxygen Losses		Constant.
		Cycle Gas Inlet Temperature	C	No Temperature transmitter.
	Norms Basis - Constants	Cycle Gas CP	kcal/kg/C	Constant.
		Cycle Gas Flow	kg/hr	No Flow transmitter.
		EO Molecular Weight	g/mol	Constant.
		Ethylene Molecular Weight	g/mol	Constant.
		CO2 Molecular Weight	g/mol	Constant.
		Oxygen Molecular Weight	g/mol	Constant.
		Stoichiometric Ratios	g/mol	Constant.
		TDI, TST vs Reactor outlet	C	No Temperature transmitter.
	Report	Budgeted Selectivity	%	No Calculation

Screen Name	Category	Manual Entry / Particulars	UoM	Reason
	Manual Entry			Available.
		Actual Selectivity	%	No Calculation Available.
		Guaranteed Selectivity	%	No Calculation Available.
		Actual Operating Hours	Hrs	Operating hours of last year (Only for reporting).
Production Target	Design Capacity	EO	TPH	As per plant design capacity provided by Licensor.
		EOE	TPH	
		MEG	TPH	
Business Demand	Production	EO	%	From Business planning team.
		EOE	%	
		MEG	%	
	By Product	CO2	MT	
Shutdown /TA activity	Shutdown /TA activity	Manual entry for Shutdown Duration		From Business planning team.
Slowdown activity	Slowdown activity	Manual entry for Slowdown Duration		From Business planning team.
Steady State Norms	Cat-Chem	CAUSTIC SODA LYE – GRADE 1	MT/MT EOE	No Calculation Available.
		ETHYLENE DI CHLORIDE (99.7%)	kg/MT EOE	
		POTASSIUM CARBONATE TECH.GRD-.02% CL	MT/MT EOE	
		CHEM CATAcarb-922	MT/MT EOE	
		CHEM PLURONIC PE 10100	kg/MT EOE	
		CHEM TRISODIUM PHOSPHATE	kg/MT EOE	
		CHEM MORPHOLENE	MT/MT EOE	
		KEM WATREAT B 70M	kg/MT	

Screen Name	Category	Manual Entry / Particulars	UoM	Reason
			EOE	
Shutdown Consumption	By Product	POLY ETHYLENE GLYCOL	MT	No Calculation Available.
		Carbon Dioxide	MT	
	Raw Material	ETHYLENE	MT	
		OXYGEN	MT	
	Utilities	HP STEAM_DIS	MT	
		MP STEAM_DIS	MT	
		LP STEAM_DIS	MT	
		POWER_DIS	kWh	
		COOLING WATER 2	km3	
		Methane Ballast	MT	
		D M WATER	M3	
		Nitrogen Gas	NM3	
		COMPRESSED AIR	NM3	
		Water	M3	
	Cat-Chem	CAUSTIC SODA LYE – GRADE 1	MT	
		ETHYLENE DI CHLORIDE (99.7%)	kg	
		POTASSIUM CARBONATE TECH.GRD-.02% CL	MT	
		CHEM CATAcarb-922	MT	
		CHEM PLURONIC PE 10100	kg	
		CHEM TRISODIUM PHOSPHATE	kg	
		CHEM MORPHOLENE	MT	
		KEM WATREAT B 70M	kg	
Plant Contribution (T-21)	Other Variable costs	STORES & SPARES other than S/D	Rs/MT	No Calculation Available.
		Shutdown Cost	Rs/MT	No Calculation Available.
		ONE TIME	Rs/MT	No Calculation Available.
		OTHER COST (S&S Allocated + RM Allocated)	Rs/MT	No Calculation Available.
Plant Contribution	Other Variable	STORES & SPARES other than S/D	Rs/MT	No Calculation Available.

Screen Name	Category	Manual Entry / Particulars	UoM	Reason
(T-22)	costs	Shutdown Cost	Rs/MT	No Calculation Available.
		ONE TIME	Rs/MT	No Calculation Available.
		OTHER COST (S&S Allocated + RM Allocated)	Rs/MT	No Calculation Available.
Turnaround report (T-19A)		Manual entry for Turnaround details for the previous years since commissioning.		No Calculation Available.